Exercise 6.1 SOLUTION

1. Find the next four terms of each sequence: i) 12,16,20,.... ii) 3,1,−1,....

Sol:

i) Sequence: 12,16,20,...
Common difference: $d = 16 - 12 = 4$
Next four terms:
$20 + 4 = 24$
$24 + 4 = 28$
$28 + 4 = 32$
$32 + 4 = 36$

ii) Sequence: 3,1,−1,...
Common difference: $d = 1 - 3 = -2$
Next four terms:
$-1 + (-2) = -3$
$-3 + (-2) = -5$
$-5 + (-2) = -7$
$-7 + (-2) = -9$

نوٹ
دیے گئے سوال میں n کی جگہ 1، 2، 3 لگائیں گے تو پہلی، دوسری اور تیسری ٹرم آجائے گی۔

2. Write down the first three terms of each of the following sequence:

Sol:

i) $a_n = 3n + 5$
For the first term $(n = 1): a_1 = 3(1) + 5 = 3 + 5 = 8$
For the second term $(n = 2): a_2 = 3(2) + 5 = 6 + 5 = 11$
For the third term $(n = 3): a_3 = 3(3) + 5 = 9 + 5 = 14$
The first three terms are $8, 11, 14$

ii) $a_{n+1} = 4a_n - 7$ and $a_1 = 3$
The first term is given $a_1 = 3$
For the second term $(n = 1): a_{1+1} = a_2 = 4a_1 - 7 = 4(3) - 7 = 12 - 7 = 5$
For the third term $(n = 2): a_{2+1} = a_3 = 4a_2 - 7 = 4(5) - 7 = 20 - 7 = 13$
The first three terms are 3,5,13

iii) $a_n = (n - 3)(n + 1)$
For the first term $(n = 1): a_1 = (1 - 3)(1 + 1) = (-2)(2) = -4$
For the second term $(n = 2): a_2 = (2 - 3)(2 + 1) = (-1)(3) = -3$
For the third term $(n = 3): a_3 = (3 - 3)(3 + 1) = (0)(4) = 0$
The first three terms are $-4, -3, 0$

iv) $a_1 = -1, a_{n+1} = \dfrac{3}{a_n + 2}$
The first term is given: $a_1 = -1$
For the second term $(n = 1): a_2 = \dfrac{3}{a_1 + 2} = \dfrac{3}{-1 + 2} = \dfrac{3}{1} = 3$
For the third term $(n = 2): a_3 = \dfrac{3}{a_2 + 2} = \dfrac{3}{3 + 2} = \dfrac{3}{5}$
The first three terms are $-1, 3, \dfrac{3}{5}$

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v) $a_n = 8 - \frac{20}{3+n}$

The first term is given: $(n=1): a_1 = 8 - \frac{20}{3+1} = 8 - \frac{20}{4} = 8 - 5 = 3$

For the second term $(n=2): a_2 = 8 - \frac{20}{3+2} = 8 - \frac{20}{5} = 8 - 4 = 4$

For the third term $(n=3): a_3 = 8 - \frac{20}{3+3} = 8 - \frac{20}{6} = 8 - \frac{10}{3} = \frac{24-10}{3} = \frac{14}{3}$

The first three terms are $3, 4, \frac{14}{3}$

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equation میں n کی جگہ 2 اور 3 لگانا ہے صرف ز دیا گیا ہے

vi) $a_1 = 1, a_{n+1} = (3a_n + 2)^2$

The first term is given: $a_1 = 1$

For the second term $(n=1): a_2 = (3a_1 + 2)^2 = (3(1)+2)^2 = (3+2)^2 = 5^2 = 25$

For the third term $(n=2): a_3 = (3a_2 + 2)^2 = (3(25)+2)^2 = (75+2)^2 = 77^2 = 5529$

The first three terms are $1, 25, 5929$

vii) $a_n = (-2n)^2$

For the first term $(n=1): a_1 = (-2(1))^2 = (-2)^2 = 4$

For the second term $(n=2): a_2 = (-2(2))^2 = (-4)^2 = 16$

For the third term $(n=3): a_3 = (-2(3))^2 = (-6)^2 = 36$

The first three terms are $4, 16, 36$

viii) $a_n = (-1)^n 7n^2$

For the first term $(n=1): a_1 = (-1)^1 .7(1)^2 = (-1).7(1) = -7$

For the second term $(n=2): a_2 = (-1)^2 .7(2)^2 = (1).7(4) = 28$

For the third term $(n=3): a_3 = (-1)^3 .7(3)^2 = (-1).7(9) = -63$

The first three terms are $-7, 28, -63$

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3. **An expression for n^{th} triangular number is $\frac{n(n+1)}{2}$. Write down the 15th triangle number. Make triangle of dots by taking $n = 5$**

Sol: The formula for the nth triangular number is given by:

$$T_n = \frac{n(n+1)}{2}$$

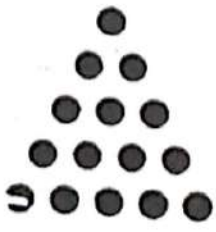
To find the 15th triangular number, we substitute $n = 15$ into the formula:

$$T_{15} = \frac{15(15+1)}{2} = \frac{15(16)}{2} = \frac{240}{2} = 120$$

15th triangular number is 120.

Now for $n = 5$; $T_5 = \frac{5(5+1)}{2} = 15$

Triangle is



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$n = 5$ لگانے سے five row والی triangle بنی ہے جس میں 15 dot

تھے اس طرح اگر $n = 15$ یعنی 15 لائن والی triangle بنائیں اسی کو

نیچے بڑھا کر تو اس میں 120 dot ہوں گے

4. **Write down the nth term of each of the following sequences:**

Sol:

i) $1,4,9,\ldots$

Notice that these are the squares of the natural numbers:

$1=1^2, 4=2^2, 9=3^2$

So, the nth term of this sequence is:

$a_n = n^2$

ii) $1, 1+2, 1+2+3, \ldots.$

The terms of this sequence are the sums of the first n natural numbers. We know that the sum of the first n natural numbers is given by the formula $\frac{n(n+1)}{2}$.

Therefore, the nth term of this sequence is:

$$a_n = 1+2+3+\ldots+n = \frac{n(n+1)}{2}$$

iii) $a_1b_1, a_2b_2, a_3b_3, \ldots.$

Following the pattern, the nth term of this sequence is simply the product of the nth terms of the two underlying sequences $\{a_n\}$ and $\{b_n\}$

$a_n = a_n b_n$

iv) $x, 2x^2, 3x^3, \ldots.$

Here

$a_1 = (1)x^1$

$a_2 = (2)x^2$

$a_3 = (3)x^3$

$\vdots \quad \vdots$

$a_n = nx^n$

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دی گئی 3 یا 4 ٹرم سے اندازہ کرنا ہے کہ کس ترتیب سے اگلی ٹرم آ رہی ہے یعنی تیسری کے بعد چوتھی کیسی ہو گی پانچویں کیسی ہو گی پھر اس کی جگہ n لگائیں گے تو nth آ جائے گی۔

v) $a_1, a_1+d, a_1+2d, \ldots$

Here

$a_1 = a_1 + 0d = a_1 + (1-1)d$

$a_2 = a_1 + d = a_1 + (2-1)d$

$a_3 = a_1 + 2d = a_1 + (3-1)d$

$\vdots \quad \vdots$

$a_n = a_1 + (n-1)d$

vi) $a_1, a_1r, a_1r^2, \ldots.$

Here

$a_1 = a_1r^0$

$a_2 = a_1r = a_1r^{(2-1)}$

$a_3 = a_1r^2 = a_1r^{(3-1)}$

$\vdots \quad \vdots \quad \vdots$

$a_n = a_1r^{n-1}$

Hence $a_n = a_1r^{n-1}$

vii) $\frac{a_1}{b_1+c_1}, \frac{2a_2}{b_2+c_2}, \frac{3a_3}{b_3+c_3}, \ldots.$

Looking at the pattern in the numerators and denominators:

The numerator of the nth term is $n.a_n$.

The denominator of the nth term is $b_n + c_n$.

Therefore, the nth term of this sequence is: $a_n = \frac{na_n}{b_n+c_n}$

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viii) $\frac{1}{a_1}, \frac{1}{a_1+d}, \frac{1}{a_1+2d}, \ldots$

Here $\frac{1}{a_1} = \frac{1}{a_1+0d} = \frac{1}{a_1+(1-1)d}$, $\frac{1}{a_2} = \frac{1}{a_1+d} = \frac{1}{a_1+(2-1)d}$

$\frac{1}{a_3} = \frac{1}{a_1+2d} = \frac{1}{a_1+(3-1)d} \ldots\ldots, \frac{1}{a_n} = \frac{1}{a_1+(2-1)d}$

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